Lucas Eduardo Contreras Delgado

Computer Science Engineering Student · Cybersecurity, Networking and Software Development

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LinkedIn · GitHub · Portfolio

Available for internship: December 2026 – February 2027 (320 hours) · on-site, hybrid or remote

Summary

Fourth-year Computer Science Engineering student at PUCV (Chile). I've already built software that real people use: scrapers that produced around 100,000 records for a risk-analysis company, a cybersecurity platform for a municipality, and a home server where I learn networking and security hands-on. I'm looking for my second internship in cybersecurity or networking, and I'm also excited about software development. I like understanding how things work under the hood, and I don't drop a problem until it's solved.

Experience

Data Automation Developer (part-time) — Mirrow · Mar 2026 – Jul 2026

- Built Python scrapers that pull company incorporations, amendments and dissolutions from Chile's Official Gazette (Diario Oficial), covering editions from around 1980 to today.
- Produced about **100,000 structured records** (JSON and PostgreSQL) used to assess whether companies are trustworthy: owners, partners, history and debts.
- Processed scanned editions with OCR as well as digital documents, automating research that used to be done by hand, one company at a time.
- Ran the pipeline with a Flask task manager on Docker, Alembic migrations, date-range processing and automatic email alerts.

Internship I — Cybersecurity & IT Support — Synthon Chile (pharmaceutical plant) · Dec 2025 – Feb 2026

- Designed and coded the prototype of an IT equipment and spare-parts inventory app in Python (Flet + SQLite) with barcode lookup, then handed it over to another intern to finish.

- Provided on-site IT support to users and equipment under pharmaceutical industry regulations and procedures (GMP).

Lead Developer — Real-client university project — Municipality of Santo Domingo, Chile · 2026

- Led the development of a citizen cybersecurity web platform: resident sign-up, risk questionnaires with scoring, learning modules, incident reports with file uploads and municipal alerts.
- Built the frontend with React + TypeScript and the backend with Node.js (Express) on Supabase (PostgreSQL), with JWT authentication and user roles.
- Applied security measures: Row Level Security, password hashing, rate limiting, file validation and HTTPS. Integrated the Google Safe Browsing API to check suspicious URLs.
- Deployed with Docker Compose (Nginx + API), gathered requirements with the municipality's IT lead and ran the handover training.

Freelance Web Developer — TunelHincado Riegomat · 2024 – 2026

- Built the company website with HTML/CSS (2024).
- Migrated it to React (2025–2026) and added a contact form that sends automatic emails (PHP).

Projects

Home lab + MineWEB · 2026 – present

- Set up my own home server (HP 250 G, 12 GB RAM) running Debian 13, Docker and CasaOS, which I manage end to end.
- Exposed it to the internet safely through a WireGuard tunnel I configured to an Oracle Cloud VPS, a firewall with only the needed ports open, and regular backups. Next step: fail2ban and network segmentation.
- On top of it I built MineWEB, a platform to rent private Minecraft servers with a self-service control panel (console, version, resources and backups). Tested with 2 servers and 4 concurrent players with no downtime.

Research: triage and waiting-line optimization and prediction · with a PUCV faculty advisor · starting 2027

- Upcoming project combining simulation and machine learning to predict demand and waiting times in emergency departments.

SPACE-NAV — 2D game · 2024

- Java game built with libGDX: progressive rounds, power-ups and an interactive tutorial, using the Strategy, Template Method and Singleton patterns.

Education & Certifications

Civil Engineering in Computer Science (Ingeniería Civil Informática) — Pontificia Universidad Católica de Valparaíso (PUCV) · 2023 – present (8th semester)

- Relevant courses: Optimization, Algorithms Fundamentals, Data Structures, Hardware & Operating Systems, Algorithm Analysis & Design.
- Also completed: Computer Networks, Parallel & Distributed Computing, Software Engineering, Web & Mobile Engineering, Databases.
- Currently taking: Cybersecurity, Machine Learning 2, Database Workshop, Requirements Engineering.

CCNA: Introduction to Networks — Cisco Networking Academy · December 2025

Skills

- **Networking & security:** OSI and TCP/IP models, IPv4/IPv6 addressing and subnetting, basic Cisco device configuration, firewalls and VPN tunnels (WireGuard), JWT and role-based access control, Row Level Security, rate limiting.
- **Systems & infrastructure:** Linux (Debian), Docker and Docker Compose, Nginx, CasaOS, Oracle Cloud, Git and GitHub.
- **Development:** Python, JavaScript, TypeScript, Java, C, SQL, PHP · React, Node.js, Express, Flask, REST APIs.
- **Data:** PostgreSQL, Supabase, MySQL, SQLite, Alembic, web scraping, OCR, ETL.

Languages & Other

- **Spanish:** native · **English:** B2 (not certified), comfortable reading technical documentation; currently taking an English course at Santander Open Academy.
- Helped with teaching-assistant sessions and the student council; took part in an internal PUCV programming competition.
- Summer jobs in sales and logistics (harvesting, loading and selling turf grass): dealing with customers and working under pressure.